

Near Net Zero Energy House

Results are in for a full year of energy generation and use at the Near Net Zero Energy house, in Colrain, Massachusetts, developed and built by Rural Development, Incorporated, (RDI) of Turners Falls, Massachusetts (see “Toward an Affordable Zero-Energy Home,” *HE* Sept/Oct ’08, p. 16). Steven Winter Associates of Norwalk, Connecticut, has been monitoring the home since its completion, with funding from DOE’s Building America program.

The 3.4 kW Northeast Solar Design Associates PV system installed at the home produced 96% of the electricity that was used during the 12-month period from June 2007 through May 2008—including a winter with lots of snow. Fifty-seven square feet of evacuated-tube solar collectors donated by American Solar Works provided 34% of the heat and hot water used for the radiant floor heat and the domestic hot water (DHW).

The three-bedroom, 1,350 ft² home was built for a total cost of \$255,872, including photovoltaics, land acquisition, and all development costs. Subsidies and donations from organizations, such as the Western Massachusetts Electric Company (WMECO) brought the purchase price down to \$199,000.

Energy efficiency measures included:

- R-43 cellulose insulation dense-packed into double-framed walls;

- R-50 cellulose blown into the attic;
- low-e windows with a U-value of 0.20 (R-5);
- R-20 insulated radiant-slab floor;
- Energy Star appliances and CFL lighting;
- on-demand propane water heater to provide auxiliary heat to the solar hot water system for the radiant floor and DHW; and
- low-flow faucets, showerheads, and dual-flush toilets.

Preconstruction modeling of the PV system predicted that it would generate 3,855 kWh over the course of a year, or 73% of total electricity consumed. In reality, the system generated only 3,373 kWh. Even so, this was sufficient to supply 96% of the family’s needs, since the three-person conservation-minded household consumed only 3,513 kWh, rather than the predicted 5,259 kWh (see Figures 1 and 2).

Results for the solar-thermal system were also impressive. Preconstruction modeling predicted that the system would offset 218 gallons of propane, providing 38% of the family’s total heat and DHW needs. Actual loads for the 12-month period were measured in 1,000 Btu (kBtu), rather than in gallons of propane, since propane is not delivered on a monthly basis. The data show that 5,746 kBtu were used for DHW and 18,122 kBtu were used for radiant floor heat, for a total load of 23,869 kBtu. The solar-thermal collectors provided



Unseen features in RDI’s Near Net Zero Energy home matter at least as much as what’s on the roof in keeping energy usage extremely low.

8,812 kBtu, or 34% of the household’s heat and hot water needs. The predicted performance was remarkably close to reality, given the very heavy snowfall last winter. RDI believes that the evacuated tubes did not shed snow as well as flat plate collectors might have. (See Figures 3 and 4 for a breakdown of the energy delivered by propane and solar and the breakdown of energy loads for heating the home and for DHW.)

Traditional heating systems are not small enough for a house with this low a heating load. Radiant floor heating was the best choice, given the site conditions and the need to build on a slab. However, the radiant heating system required two pumps to run continuously throughout the winter, thereby accounting for a large portion of the electric load. RDI’s next project, the Wisdom Way Solar Village, in Greenfield, Massachusetts, builds on the realization that ducted central heating was not really necessary in the Colrain house due to the low heat load.

EDUCATION MAKES A DIFFERENCE

Education really does make a difference when it comes to pushing the envelope and achieving ever better home performance. Anne Perkins, RDI’s director of Home Ownership programs, was already building affordable homes that went beyond Energy Star in their energy efficiency when she attended the Northeast Sustainable Energy Association (NESEA) BuildingEnergy conference in Boston in 2006 and learned about Passive House (or Passivhaus) design. (See “First U.S. Retrofit to Passive House Standards,” p. 25.) Architect Bill Austin of Austin Design in Colrain, Massachusetts had recently offered to design a home for RDI pro bono and had begun to do so. While she was still at the conference, Anne ran into

Robb Aldrich of Steven Winter Associates and excitedly discussed the idea of building a near net zero energy affordable home. Robb thought DOE’s Building America might want to be involved in such a project. Back in western Massachusetts, Anne called Bill and asked him if his design could be changed to a new near net zero energy plan. His response was enthusiastic, and they recruited others to be part of an integrated design team. The results were, first, the Colrain home and now the Wisdom Way Solar Village. NESEA’s next BuildingEnergy conference takes place in Boston, March 10–12, 2009. There are sure to be more ideas raised there than can be put into action.



ADAM GRANDIN

The Colrain house sits snugly in its tranquil rural setting as autumn mists rise to the South. This view as one approaches from the North shows how few trees had to be cleared to maximize solar exposure.

Therefore each house in the Solar Village will rely on passive-solar heat gain, with a backup natural gas room heater installed in the central living space.

It is important to emphasize how much the family's habits affect the amount of energy used in any house, which, in turn, will effect the percentages generated through solar. At the same time note that living in an energy-conscious way imposes no hardship on the Colrain family. They do not take extreme conservation measures, nor do they need to. In their kitchen, they have an electric dishwasher, a toaster oven, an electric kettle, and other small appliances. They have a propane oven and range for cooking and baking. They use a vacuum cleaner, like most of us. Clothes are washed in an electric washing machine, hung outside to dry on sunny days, and dried in an electric dryer when the sun isn't shining. For entertainment, they have two televisions for watching movies, two computers, and several radios. These devices are plugged into power strips, which are turned off when not in use to avoid phantom loads on the electrical service. All of the lightbulbs are CFLs. There is no air conditioning; superinsulated walls keep the house comfortable in summer as well as in winter.

RDI was awarded LEED Gold certification for the Colrain house, and it is an Energy Star-certified home (see "Education Makes a Difference"). On the HERS index, the house scored 21—with 0 being the best possible rating and 100 being the typical new code-built home today. The scale goes as high as 150 for the average American home. A home must score under 85 in order to be Energy Star certified. Anne Perkins, Robb Aldrich, Bill Austin, and the rest of RDI's integrated design team have designed the Wisdom Way Solar Village homes, due for completion in late 2009, to have HERS scores as low as 18. Watch *Home Energy* for more info on the project. **He**

—Wendy Forbes

Wendy Forbes is the Home Ownership Programs assistant at Rural Development, Incorporated, (RDI) in Turners Falls, Massachusetts.

affordable housing

Electricity Consumed, Generated by PV

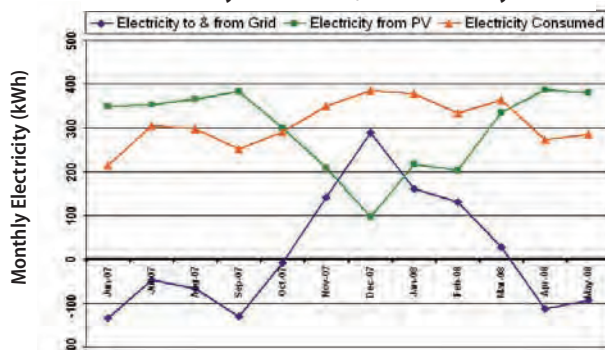


Figure 1. Electricity supplied from the grid appears as a positive number in the above chart. Electricity sent to the grid from the PV system appears as a negative number.

Electric Loads

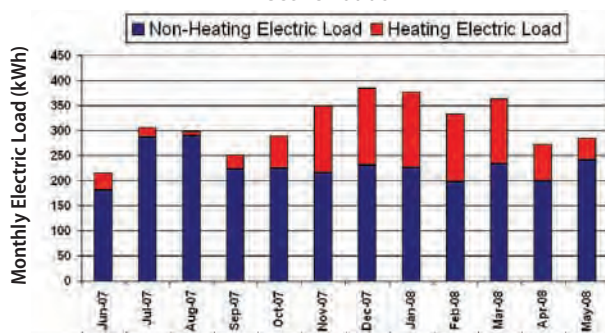


Figure 2. Electricity generated by PV panels met 96% of the home's load.

Space and Water Heating Energy Supply

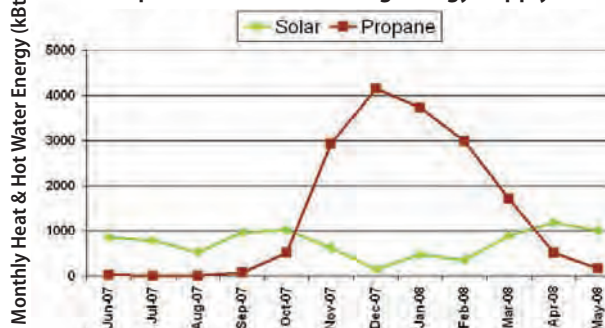


Figure 3. A backup propane water heater met most of the heat and hot water needs in the winter.

Space and Hot Water Loads

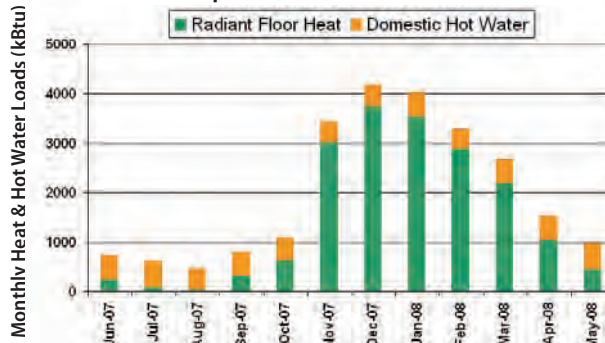


Figure 4. The solar-thermal collectors provided 8,812 kBtu, or 34% of the household's heat and hot water needs.